

MOSAIC: Multimodal Perceptual Cue Fusion for Full-Reference Point Cloud Quality Assessment

Jiaxin Li, Tong Chen, Zehong Li, Dandan Ding, and Zhan Ma

Abstract—Motivated by David Marr’s three-level account of vision, we formulate full-reference point cloud quality assessment (FR-PCQA) as a multimodal perceptual inference problem. At the computational level, the goal is to predict perceived quality from reference-distorted discrepancies. At the representational level, quality is described by complementary cues of 3D geometry, 3D color, and 2D visual saliency. At the implementational level, these cues are realized through point-cloud feature extraction and multi-view saliency projection. Based on this formulation, we propose MOSAIC, a multimodal perceptual cue fusion framework for FR-PCQA. Instead of aggregating heterogeneous descriptors in a flat feature pool, MOSAIC first measures and fuses geometry and color fidelity in the native 3D domain as intrinsic fidelity cues, and then incorporates projected saliency consistency to model attention-related perceptual importance. A two-stage adaptive fusion strategy integrates these cues to predict the final perceptual fidelity score without MOS-supervised training or regression-based score integration. Experiments on five public PCQA databases show that MOSAIC achieves the best training-free performance on WPC and LS-PCQA, with PLCC/SROCC values of 0.844/0.834 and 0.622/0.598, respectively, while maintaining competitive accuracy across datasets and distortion types.

Index Terms—Point Cloud Quality Assessment, Multimodal Fusion, 3D Geometry, 3D Color, 2D Saliency

I. INTRODUCTION

With the growing popularity of immersive applications such as Virtual/Augmented/eXtended Reality (VR/AR/XR), point clouds have attracted significant attention for their simple yet efficient representation of 3D objects and scenes. However, along with acquisition, compression, transmission, and rendering, associated processing distortions are inevitably introduced, leading to visual impairments and reduced Quality of Experience (QoE). Therefore, point cloud quality assessment (PCQA), which aims to develop an effective analytical model to accurately predict subjective quality metrics such as MOS (Mean Opinion Score), is crucial for quantitatively monitoring and optimizing system-level QoE for underlying services.

Depending on the availability of the original reference, PCQA methods can be categorized as full-reference (FR), reduced-reference (RR), or no-reference (NR). Leveraging complete reference information, FR-PCQA methods generally achieve more stable and reliable performance [1].

However, the advantage of complete references is still constrained by how reference-distorted differences are represented and pooled. Direct geometry and color metrics are efficient, yet they mainly measure point-wise or low-level

local discrepancies and often treat geometry and appearance separately [2]–[4]. Handcrafted structural metrics introduce local neighborhoods, graph representations, multiscale analysis, or saliency-assisted pooling [1], [5]–[8], but heterogeneous cues are usually aggregated within a single quality layer or remain dominated by intrinsic 3D evidence. Regression-based and deep-learning FR methods can improve correlation with subjective scores by learning score mappings or visual features [9]–[13]; nevertheless, their performance may be tied to subjective-label fitting, data splits, rendered-view settings, backbone choices, or prompting and fine-tuning strategies. These observations suggest that the remaining limitation is not the lack of reference information, but the lack of an explicit and interpretable organization of heterogeneous perceptual cues.

This issue is also consistent with how humans perceive 3D visual content. Human vision recovers scene layout and surface structure from image measurements rather than directly observing 3D geometry [14], and psychophysical studies show that depth and surface judgments integrate multiple cues, such as stereo and texture, under perceptual uncertainty [15], [16]. Meanwhile, point clouds are typically inspected through rendered views, where visibility, projected appearance, and image-plane visual attention influence perceived quality [17]–[19]. Therefore, point-cloud quality perception is naturally a mixed 3D–2D process: intrinsic 3D structure and attributes provide object-level fidelity, whereas 2D projections determine view-dependent visibility and attention. This leads to the central question of this work: how can complete reference information be organized into a training-free and interpretable fusion of 3D structural, 3D appearance, and 2D attention-related cues?

To answer this question, we revisit FR-PCQA from Marr’s three-level perspective of vision [14]. At the computational level, the task is to infer perceived quality from the discrepancy between a reference and a distorted 3D representation. At the representational level, this discrepancy should be described by perceptually meaningful cues, including geometric structure, appearance consistency, and attention-related visual importance. At the implementational level, these cues can be realized through measurable descriptors extracted from point clouds and their projected views. This perspective motivates MOSAIC, a Multimodal Perceptual Cue Fusion framework that integrates intrinsic 3D cues with projected 2D saliency cues. Specifically, we use established local geometric descriptors not as a standalone novelty, but as interpretable structural fidelity cues within a broader perceptual representation. These cues are complemented by density, color, and projected

saliency information, and their modality-wise fidelity scores are then combined through a two-stage fusion process that first integrates intrinsic 3D structure and appearance and then incorporates attention-related saliency fidelity.

The main contributions are summarized as follows:

- We formulate FR-PCQA from Marr’s three-level perspective, treating point cloud quality prediction as a multimodal perceptual inference problem, and propose MOSAIC, a multimodal perceptual cue-fusion framework that repurposes interpretable 3D structural descriptors together with 3D color appearance and 2D saliency-based visual attention.
- We introduce a two-stage adaptive fusion strategy that first integrates intrinsic 3D fidelity cues and then incorporates attention-related saliency consistency without MOS-supervised model fitting.
- Extensive experiments on five public PCQA databases demonstrate the robustness and generalization ability of MOSAIC, including the best training-free performance on WPC and LS-PCQA.

II. RELATED WORK

Local structure-aware FR-PCQA. Early FR-PCQA metrics mainly measure direct point-wise discrepancies between the reference and distorted point clouds. Point-to-point (p2po) [2] and point-to-plane (p2pl) [3] distances quantify geometric errors, while luminance or color PSNR measures attribute distortion [4]. These metrics are simple and efficient, but geometry and color are treated largely as separate low-level signals. Later training-free metrics moved from isolated point errors to local structural comparison. PCQM [5] extracts local geometric and color descriptors from matched point neighborhoods and combines a selected set of curvature- and lightness-related features into a handcrafted full-reference score. PointSSIM [20] extends structural similarity to point clouds by computing statistical similarities of geometry, normal, curvature, or color features within local neighborhoods. GraphSIM [6] constructs local graphs around key points, treats color attributes as graph signals, and combines graph-based color similarity with geometric weighting; MS-GraphSIM [7] further evaluates graph similarity at multiple neighborhood radii. MPED [21] describes local distortions through multi-scale potential energy discrepancy. More recently, PQI [1] refined the same neighbor-to-neighbor paradigm by selecting scale-wise key points, using mild neighborhoods, and measuring geometry and attribute similarity at multiple scales. Thus, PQI should be viewed as a multiscale key-point and neighborhood refinement of the local-structure lineage rather than as a method of the same type as TCDM or PHM. Saliency has also been explored in PQSM [8], which generates 3D saliency maps through projection, derives geometry, color, and saliency discrepancies in local neighborhoods, and pools local quality scores with saliency guidance. Overall, this family improves over raw point-wise distances by better describing local structure, but the final score is still mainly obtained by handcrafted single-layer pooling over intrinsic geometry, color, or local saliency evidence.

Perceptual-hypothesis-driven FR-PCQA. Beyond refining key points and neighborhoods, several recent training-free metrics define a higher-level perceptual or computational hypothesis for quality prediction. TCDM [22] formulates quality assessment through transformational complexity: it partitions the reference and distorted point clouds into local 3D Voronoi patch pairs and uses a space-aware vector autoregressive model to estimate self-complexity, transformational complexity, and prediction-related terms. PHM [23] adopts a perception-guided hybrid strategy based on distortion degree: a visible-difference branch handles high-quality samples using texture-complexity-compensated PSNR_Y, while an appearance-degradation branch handles lower-quality samples using local graph construction, geometry signal variation, spectral graph wavelet coefficients, and weighted co-occurrence statistics. These methods differ from PCQM, PointSSIM, GraphSIM, MS-GraphSIM, MPED, and PQI because their main novelty is not only a better local descriptor, but a specific perceptual organization of quality evidence: TCDM tests whether recovery complexity is a proxy for perceptual degradation, whereas PHM tests whether different distortion levels require different HVS-inspired strategies. However, each hypothesis mainly organizes quality evidence within one dominant axis, such as local transformational complexity or distortion-level-dependent graph/spectral analysis. MOSAIC follows the same principle of explicit hypothesis-driven metric design, but its hypothesis is different: perceived quality is better modeled when intrinsic 3D structure and appearance are first integrated and then modulated by view-dependent saliency.

Regression-based FR-PCQA. Another line uses handcrafted or graph-derived descriptors as input features and learns a mapping to subjective scores. PointPCA [9] builds objective quality descriptors from PCA-based local neighborhood analysis and uses these descriptors for quality regression. This strategy can improve correlation with MOS under a predefined training/testing protocol, but it requires subjective-label supervision and split-dependent fitting. FRSVR [10] explicitly targets efficient score regression: it computes five full-reference metrics covering geometry distortion, color distortion, and point-count changes, and then integrates them using support vector regression (SVR). SGW-PCQA+ [11] further uses spectral graph wavelet representations with supervised score integration. More recently, the Feature Selection Model (FSM) [24] benchmarks features from several representative FR metrics, applies recursive feature elimination with SVR or ridge regression, and combines selected features from PCQM, MS-GraphSIM, and PSNR MSE D2 through ridge regression. Regression-based methods are useful high-performing references because they learn dataset-specific score mappings, but this also means their numerical performance is not strictly comparable with training-free metrics unless training protocols, data splits, and subjective-label usage are carefully controlled.

Projection-based and deep-learning FR-PCQA. Projection-based PCQA renders point clouds into multiple 2D views and reuses mature image-quality or visual-representation tools [18], [19], [25]–[27]. In learning-based FR-PCQA, Tliba

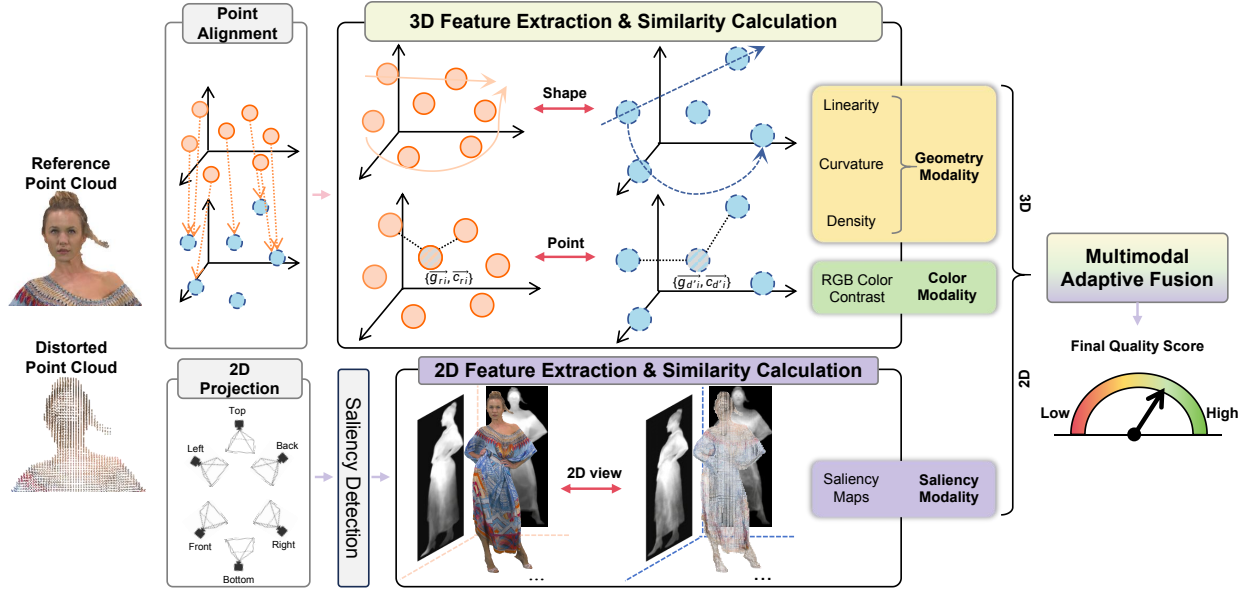


Fig. 1. Overview of the proposed MOSAIC framework from Marr's three-level perspective. The computational goal is to predict perceptual quality from discrepancies between reference and distorted point clouds. The representational level comprises geometric, color, and saliency cues. The implementational level realizes these cues through 3D feature extraction, 2D saliency estimation, and two-stage adaptive fusion.

et al. [28] extract local deep features from reference and distorted point-cloud patches and measure visual degradation through cross-correlation between the learned embeddings. Simple projection-based deep baselines [27] render reference and distorted point clouds into cube-like or multi-view images, extract feature maps using modern backbones such as ConvNeXt V2 or Swin Transformer, and use feature similarity or learned pooling to predict quality. CT-PCQA [12] combines CNN and Transformer backbones on projected views and supports both full-reference and no-reference prediction settings. More recently, MLLM-based FR-PCQA [13] has investigated rendered-view prompting, response similarity, relative-quality response scores, fine-tuned absolute-quality scoring, and SVR-based score integration with conventional FR metrics. Adjacent MLLM-based NR-PCQA work, such as LMM-PCQA [29], converts quality labels into textual quality levels and combines LMM outputs with multiscale structural features, further illustrating the broader trend toward multimodal quality representations.

These learning-based models demonstrate the value of high-level visual representations, but they introduce practical dependencies that are less desirable for a deterministic objective metric: subjective-label training, backbone choice, rendering configuration, prompt design, fine-tuning strategy, and heavier inference cost. Therefore, MOSAIC does not aim to replace high-capacity learned regressors when abundant labeled data are available. Instead, it targets a complementary setting: a training-free, interpretable, and computationally moderate FR-PCQA metric that jointly measures native 3D geometry fidelity, 3D color fidelity, and projected saliency fidelity within one multimodal cue-fusion framework with a two-stage organization.

III. METHOD

Guided by Marr's theory, MOSAIC extracts cues of 3D structure, color appearance, and visual saliency and fuses them for FR-PCQA. The general framework is shown in Fig. 1. For a point cloud $\mathbf{P} = \{\vec{p}_1, \vec{p}_2, \dots, \vec{p}_N\} \in \mathbb{R}^{N \times 6}$, each point $\vec{p}_i$ is represented as a vector comprising 3D coordinates $\vec{g}_i$ and three-channel color attributes $\vec{c}_i$, i.e., $\vec{p}_i = \{\vec{g}_i, \vec{c}_i\} = \{(x_i, y_i, z_i), (r_i, g_i, b_i)\}$, $i \in [1, N]$, with a total of N points.

A. Point Alignment

To align the reference and distorted point clouds ($\mathbf{P}_r \in \mathbb{R}^{N \times 6}$ and $\mathbf{P}_d \in \mathbb{R}^{M \times 6}$), each point in $\mathbf{P}_r$ is treated as a keypoint. Its corresponding point in $\mathbf{P}_d$ is identified via a nearest-neighbor search based on Euclidean distance in 3D space. These matched points collectively form the aligned distorted point cloud $\mathbf{P}_{d'} \in \mathbb{R}^{N \times 6}$ as the correspondence of $\mathbf{P}_r$.

B. Multimodal Perceptual Representation

1) *3D Structural Representation via Geometry Features:*
Local shape feature. Geometric distortions degrade local spatial structures such as surfaces and edges, thereby affecting perceptual quality. To characterize local geometry, eigenvalue-based descriptors are extracted following prior works [9], [30], [31]. The purpose of this branch is therefore not to introduce a new local shape descriptor, but to construct an interpretable 3D structural cue that can be consistently compared between reference and distorted point clouds and later fused with color and saliency cues in the proposed cue-fusion framework.

For each point $\vec{p}_i$, a local neighborhood $\{\vec{g}_i^k\}_{k=1}^{K_g}$ is constructed using K_g -nearest neighbors. The corresponding covariance matrix is computed as

$$\text{Cov}_i = \frac{1}{K_g} \sum_{k=1}^{K_g} (\vec{g}_i^k - \vec{g}_i) (\vec{g}_i^k - \vec{g}_i)^T. \quad (1)$$

Eigen-decomposition of Cov_i yields three eigenvalues $\lambda_1 > \lambda_2 > \lambda_3$. Based on them, the local curvature $G_i^{(1)}$ and linearity $G_i^{(2)}$ are defined as

$$G_i^{(1)} = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3}, \quad G_i^{(2)} = \frac{\lambda_1 - \lambda_2}{\lambda_1}. \quad (2)$$

The same descriptors are extracted from both the reference and resampled distorted point clouds, denoted as $\{G_{ri}^{(1)}, G_{ri}^{(2)}\}$ and $\{G_{d'i}^{(1)}, G_{d'i}^{(2)}\}$, respectively.

Local point density feature. Geometric downsampling or Gaussian noise often results in changes to point density and point position. To quantify this property, Euclidean distances are computed for each point and its K_p nearest neighbors. Then the average distance is used as the local point density feature for both reference and distorted point clouds, i.e.,

$$\begin{aligned} G_{ri}^{(3)} &= \frac{1}{K_p} \sum_{k=1}^{K_p} \|\vec{g}_{ri}^k - \vec{g}_{ri}\|_2^2, \\ G_{d'i}^{(3)} &= \frac{1}{K_p} \sum_{k=1}^{K_p} \|\vec{g}_{d'i}^k - \vec{g}_{d'i}\|_2^2. \end{aligned} \quad (3)$$

$\vec{g}_{ri}$ and $\vec{g}_{d'i}$ are the coordinates of the i -th point, and $\vec{g}_{ri}^k$ and $\vec{g}_{d'i}^k$ represent the coordinates of the k -th nearest neighbor of the i -th point; K_p is the number of nearest neighbors considered for computing the local density. Local density captures fine-grained variations in the point cloud by characterizing the distribution of points in local neighborhoods. Together with curvature and linearity, this density term makes the structural cue sensitive to both local surface-shape changes and sampling-distribution changes, which are often affected differently by compression, downsampling, and noise.

2) *Appearance Representation Using Color Features:* This work adopts the native RGB color space. Although alternative spaces (e.g., YUV or LAB) have been explored in prior studies [5], [10], RGB directly represents the color attributes stored in many point-cloud datasets and avoids additional assumptions introduced by color-space conversion. As reported in Table VI, RGB also achieves the best overall performance within our framework. Motivated by the fact that human perception is sensitive to local chromatic contrast, a local RGB residual is extracted as the color feature for each point $\vec{p}_i$:

$$\mathbf{C}_{ri} = [C_{ri}^{(1)}, C_{ri}^{(2)}, C_{ri}^{(3)}] = \frac{1}{K_c} \sum_{k=1}^{K_c} (\vec{c}_{ri}^k - \vec{c}_{ri}), \quad (4)$$

$$\mathbf{C}_{d'i} = [C_{d'i}^{(1)}, C_{d'i}^{(2)}, C_{d'i}^{(3)}] = \frac{1}{K_c} \sum_{k=1}^{K_c} (\vec{c}_{d'i}^k - \vec{c}_{d'i}), \quad (5)$$

where $\mathbf{C}_{ri}$ and $\mathbf{C}_{d'i}$ denote the color features composed of R, G, B components for reference and the aligned distorted point cloud, respectively; $\vec{c}_{ri}$ and $\vec{c}_{d'i}$ represent the color of the i -th point, while $\vec{c}_{ri}^k$ and $\vec{c}_{d'i}^k$ correspond to the colors of its

k -th nearest neighbor. K_c denotes the number of neighboring points used to evaluate local color contrast.

3) *Attention-related Saliency Representation:* Introducing artifacts can alter visual saliency patterns, and the human visual system is sensitive to distortions occurring in salient regions. Therefore, visual saliency plays a crucial role in perceptual quality evaluation [32].

Following [18], we first project both reference and distorted point clouds onto six perpendicular views, which cover most visible content from canonical directions. Saliency maps are then extracted from each projected view using the established image saliency detection algorithm [33], resulting in saliency representations S_r^v and S_d^v for reference and distorted point clouds, respectively, where $v \in \{1, \dots, 6\}$ denotes the view index.

We use projected 2D saliency for two reasons. First, human observers typically perceive rendered point clouds through image-like views, making view-based saliency a natural approximation of visual attention. Second, extracting saliency in the projected domain avoids the ambiguity of defining attention directly on irregular 3D point sets.

Rather than using saliency only as a spatial weighting map, we treat it as an independent modality because distortions may alter the saliency distribution itself, which provides complementary information beyond geometry and color discrepancies.

C. Modality-wise Perceptual Fidelity Measurement

Next, we compute point-to-point similarity between the reference and aligned distorted point clouds for each geometry and color feature. The final geometry and color fidelity scores F_G and F_C are obtained by averaging over N points and three features:

$$F_G = \frac{1}{3} \frac{1}{N} \sum_{l=1}^3 \sum_{i=1}^N \frac{2G_{ri}^{(l)} G_{d'i}^{(l)} + T}{(G_{ri}^{(l)})^2 + (G_{d'i}^{(l)})^2 + T}, \quad (6)$$

$$F_C = \frac{1}{3} \frac{1}{N} \sum_{l=1}^3 \sum_{i=1}^N \frac{2C_{ri}^{(l)} C_{d'i}^{(l)} + T}{(C_{ri}^{(l)})^2 + (C_{d'i}^{(l)})^2 + T}, \quad (7)$$

where T is a small constant used to avoid numerical instability.

For the saliency modality, we calculate the structural similarity index (SSIM) [34] for each view pair (S_r^v, S_d^v) , $v \in \{1, \dots, 6\}$. Following [32], the view scores are averaged to obtain the final saliency fidelity score F_S :

$$F_S = \frac{1}{6} \sum_{v=1}^6 \text{SSIM}(S_r^v, S_d^v). \quad (8)$$

D. Two-stage Perceptual Cue Fusion

After obtaining modality-wise fidelity scores, we adopt a two-stage fusion strategy motivated by the layered organization of visual representation. Geometry and color describe intrinsic properties of the 3D object: geometry captures structural fidelity, while color captures appearance fidelity. These two cues are therefore first integrated to form an intrinsic 3D fidelity representation. Saliency, derived from projected views,

reflects the attentional importance of visible regions and is subsequently used to modulate the intrinsic representation. This design follows the intuition that perceptual quality is determined not only by the amount of physical degradation, but also by whether visually important regions are affected.

$$F_{3D} = \frac{w_{F_G} F_G + w_{F_C} F_C}{w_{F_G} + w_{F_C} + \epsilon}, \quad (9)$$

$$F = \frac{w_{F_{3D}} F_{3D} + w_{F_S} F_S}{w_{F_{3D}} + w_{F_S} + \epsilon}, \quad (10)$$

where w_{F_G} , w_{F_C} , $w_{F_{3D}}$ and w_{F_S} denote the weights associated with geometry, color, fused 3D, and saliency fidelity scores, respectively; F_{3D} represents the fused intrinsic 3D fidelity score, F denotes the final perceptual fidelity score, and ϵ is a small constant.

Perceptual studies on multisensory cue integration indicate that the human visual system does not linearly average multiple cues. Instead, their contributions are modulated in a bounded and nonlinear fashion [35]. Moreover, human visual responses to stimulus intensity often exhibit saturating nonlinear characteristics that can be well approximated by sigmoid-like functions [36].

Motivated by these observations, we employ a sigmoid function to obtain modality weights:

$$w_{F_m} = \frac{1}{1 + \exp(-\alpha F_m)}, \quad (11)$$

$$F_m \in \{F_G, F_C, F_{3D}, F_S\}.$$

where α controls the slope of the nonlinear mapping and is set to 1 in this work to balance adaptivity and stability. This formulation ensures that the weights vary smoothly with cue scores and remain within a fixed and bounded range.

IV. EXPERIMENTS

A. Experimental Settings

Datasets: We evaluate MOSAIC on five publicly accessible PCQA datasets, including SJTU [18], WPC [19], ICIP [37], LS-PCQA [38], and MJ-PCCD [39]. These datasets cover diverse distortion types and content characteristics, providing comprehensive benchmarks for evaluating prediction accuracy and generalization.

The **SJTU dataset** [18] comprises 9 reference point clouds and 378 distorted versions generated using seven distortion types, including octree compression, color noise, geometry Gaussian noise, downsampling, and their combinations, applied at six distortion levels. The **WPC dataset** [19] consists of 20 reference point clouds, each subjected to five types of distortions, including Gaussian noise, downsampling, and compression noise generated by MPEG G-PCC (Trisoup), G-PCC (Octree), and V-PCC, producing 740 distorted samples. The **ICIP dataset** [37] contains six human-body point clouds compressed using MPEG G-PCC (Octree), G-PCC (Trisoup), and V-PCC codecs at five bitrates, resulting in 90 impaired samples. The **LS-PCQA dataset** [38] includes 930 distorted point clouds created from 85 reference point clouds using 31 distortion types, where Part I is annotated with MOSs. The **MJ-PCCD dataset** [39] includes six reference point clouds

and 213 distorted samples generated using G-PCC, V-PCC, and learning-based JPEG Pleno standard codecs.

Implementation Details: For neighborhood construction, $K_g = 100$, $K_p = 10$, and $K_c = 10$ are used for geometry (Eq. 1), point density (Eq. 3), and color feature (Eqs. 4, 5) extraction, respectively. To ensure computational tractability while capturing complex spatial structures, we adopt the KNN strategy rather than a radius-based neighborhood search (see Sec. IV-B1), using a relatively large fixed neighborhood size $K_g = 100$ for local shape feature extraction, following prior studies [30], [31]. This large K_g allows for robust representation of structural patterns. Complementarily, local point density is incorporated to model spacing variations, where a small K_p focuses on fine-scale point spacing changes and improves sensitivity to local geometry distortions. A compact color neighborhood size K_c is adopted to preserve local chromatic contrast while reducing oversmoothing across heterogeneous regions. $\alpha = 1$ in Eq. 11 is used for balanced adaptive fusion.

Evaluation Criteria: Following the common PCQA evaluation protocol, prediction scores are first mapped to MOS values using a five-parameter logistic regression function [40]. We then report Pearson Linear Correlation Coefficient (PLCC), Spearman Rank Correlation Coefficient (SROCC), and root mean square error (RMSE), which measure prediction accuracy, monotonicity, and error, respectively. Higher PLCC and SROCC and lower RMSE indicate better performance.

Evaluation Protocol: For the two regression-based methods, PointPCA [9] and FRSVR [10], we follow their original testing protocols. PointPCA adopts an 80%/20% split for training and testing, respectively, where all possible splits under this ratio are exhaustively evaluated. For FRSVR, following the settings reported in Table 1 of their paper, the ICIP and MJ-PCCD datasets are split into three partitions and the WPC dataset into five partitions, with each partition alternately used for testing and the remainder for training. After all iterations are completed, the prediction scores of all test samples are concatenated, and the evaluation metrics are computed globally over the entire dataset. For all methods, to address scale differences between predicted scores and MOS values, the logistic function recommended by the ITU VQEG (Video Quality Expert Group) [40] is applied:

$$\hat{y} = \beta_1 \left(0.5 - \frac{1}{1 + e^{\beta_2(y - \beta_3)}} \right) + \beta_4 y + \beta_5, \quad (12)$$

where y represents the predicted quality score, and $\beta_1, \dots, \beta_5$ are optimized by minimizing the squared errors between the mapped scores $\hat{y}$ and the subjective MOS values.

B. Performance Evaluation

We compare MOSAIC with several widely used training-free FR-PCQA metrics: MSE-p2point (M-p2po) [2], MSE-p2plane (M-p2pl) [3], PSNRyuv [4], PCQM [5], PointSSIM [20], GraphSIM [6], MS-GraphSIM [7], MPED [21], PQI [1], TCDM [22], and PHM [23]. Regression-based and learning-based FR-PCQA methods are summarized separately in Table II, because their reported

TABLE I

PERFORMANCE EVALUATION OF TRAINING-FREE FR-PCQA METRICS ON DIFFERENT DATASETS. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN **RED** AND **GREEN**. A DASH INDICATES THAT THE RESULT IS UNAVAILABLE OR NOT DIRECTLY REPORTED UNDER THE CORRESPONDING DATASET SETTING. REPORTED RESULTS ARE ROUNDED AS IN THE CORRESPONDING REFERENCES OR BENCHMARK TABLES.

	SJTU [18]			WPC [19]			ICIP [37]			LS-PCQA [38]			MJ-PCCD [39]		
	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
M-p2po [2]	0.728	0.801	1.717	0.578	0.566	18.714	0.947	0.935	0.365	0.398	0.185	0.763	0.434	0.314	1.054
M-p2pl [3]	0.652	0.661	1.893	0.454	0.446	20.422	0.954	0.948	0.341	0.351	0.166	0.779	0.396	0.370	1.074
PSNRyuv [4]	0.649	0.646	1.847	0.551	0.536	19.132	0.868	0.868	0.564	0.390	0.386	0.766	0.416	0.310	1.064
PCQM [5]	0.860	0.847	1.237	0.750	0.743	15.164	0.845	0.866	0.608	0.404	0.362	0.761	0.581	0.442	0.953
PointSSIM [20]	0.846	0.844	1.295	0.551	0.541	19.132	0.687	0.621	0.825	0.248	0.149	0.806	0.491	0.431	1.019
GraphSIM [6]	0.856	0.842	1.256	0.687	0.678	16.649	0.906	0.889	0.481	0.252	0.243	0.805	0.574	0.546	0.958
MS-GraphSIM [7]	0.899	0.890	1.031	0.709	0.704	16.172	0.482	0.021	1.001	0.327	0.307	0.786	–	–	–
MPED [21]	0.901	0.892	1.053	0.626	0.627	17.874	0.958	0.953	0.328	0.540	0.501	0.700	0.534	0.440	0.990
PQI [1]	0.912	0.906	0.996	0.813	0.808	13.364	0.966	0.955	0.292	0.437	0.379	0.748	0.582	0.481	0.952
TCDM [22]	0.930	0.910	0.895	0.806	0.804	13.557	0.942	0.935	0.383	0.335	0.325	0.784	0.885	0.837	0.546
PHM [23]	0.906	0.888	1.025	0.838	0.833	12.510	0.969	0.958	0.280	0.546	0.500	0.697	0.780	0.690	0.733
MOSAIC	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651

TABLE II

REPORTED PERFORMANCE OF SUPERVISED REGRESSION AND LEARNING-BASED FR-PCQA REFERENCES UNDER THEIR ORIGINAL PROTOCOLS. THESE RESULTS ARE LISTED SEPARATELY FROM TABLE I BECAUSE THEY INVOLVE SUBJECTIVE-LABEL FITTING, LEARNED BACKBONES, PROMPT-BASED INFERENCE, OR CHALLENGE-SPECIFIC EVALUATION SETTINGS.

Method	Learning / integration strategy	Reported results	Protocol note
PointPCA [9]	PCA-based local descriptors with quality regression	SJTU: 0.909/0.879/1.012; WPC: 0.891/0.885/10.202	PLCC/SROCC/RMSE under the original split protocol.
FRSVR [10]	Five FR distortion measures integrated by SVR	WPC: 0.811/0.796/13.428; ICIP: 0.956/0.935/0.335; MJ-PCCD: 0.709/0.620/0.825	PLCC/SROCC/RMSE under the original split protocol.
SGW-PCQA+ [11]	Spectral graph wavelet features plus SVR-based score integration	WPC: 0.873/0.869; BASICS: 0.924/0.888	PLCC/SROCC reported in the original paper.
FR-MLLM [13]	MLLM response scores combined with FR metrics through SVR	WPC: 0.889/0.881; SJTU: 0.935/0.915; BASICS: 0.926/0.895	PLCC/SROCC; uses MLLM inference, fine-tuning, and SVR.
Projection-based deep baseline [27]	ConvNeXt/Swin features with FC quality regression	BASICS FR broad-range: 0.8706/0.8955; FR high-range: 0.6296/0.5917	PLCC/SROCC in ICIP 2023 PCVQA challenge tracks.
CT-PCQA (FR) [12]	ConvNeXt/Swin projection features with FC quality regression	SJTU: 0.929/0.910/0.861; WPC: 0.881/0.878/10.815	PLCC/SROCC/RMSE under content-separated cross-validation.

scores can depend on training data, split protocols, backbone choices, prompt design, or supervised score integration.

According to Table I, MOSAIC demonstrates consistently strong performance across all datasets. TCDM achieves the best results on SJTU and MJ-PCCD by modeling transformational complexity, while PHM performs particularly well on ICIP through its perception-guided hybrid strategy. MOSAIC obtains the best overall performance on WPC and LS-PCQA with (PLCC / SROCC / RMSE) values of (0.844 / 0.834 / 12.292) and (0.622 / 0.598 / 0.652), respectively. On MJ-PCCD, MOSAIC ranks second after TCDM and clearly improves over PHM and PQI. These results indicate that the proposed multimodal perceptual cue fusion is not tailored to a single dataset or distortion family, but provides balanced generalization across heterogeneous point cloud databases.

Time Complexity. We evaluate the computational efficiency of several FR-PCQA methods on the SJTU dataset. Since these methods are implemented in different software environments (C/C++, MATLAB, and Python), the reported runtime is provided for reference rather than a strictly fair comparison. Table III presents the average runtime per distorted point cloud. PCQM, PointSSIM, PQI, and TCDM are relatively efficient, whereas PHM has the largest runtime due to graph

TABLE III
RUNTIME COMPARISON ACROSS DIFFERENT METHODS.

Method	PSNRyuv	PCQM	PointSSIM	GraphSIM	MPED	PQI	TCDM	PHM	MOSAIC
Time (s)	10.30	3.55	4.24	11.68	7.60	4.71	5.22	14.96	6.41

TABLE IV
ABLATION STUDY OF DIFFERENT FUSION STRATEGIES.

Scheme	Fusion	WPC			LS-PCQA		
		PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
Only 3D	Adaptive	0.793	0.775	13.973	0.523	0.493	0.709
	Uniform	0.835	0.823	12.622	0.564	0.533	0.687
One-stage	Adaptive	0.836	0.825	12.573	0.586	0.560	0.674
	Uniform	0.844	0.833	12.296	0.592	0.566	0.671
Two-stage	Adaptive	0.844	0.834	12.292	0.622	0.598	0.652

construction, graph-domain feature analysis, and spectral operations. MOSAIC requires 6.41 seconds on average, which is lower than PSNRyuv, GraphSIM, MPED, and PHM, while maintaining strong cross-dataset prediction performance.

Ablation. To confirm the complementarity of the added 2D saliency modality, we first evaluate a baseline that relies solely on 3D fidelity cues. Only controlled variants of MOSAIC are treated as ablations; external supervised methods such

TABLE V
ABLATION STUDY ON NEIGHBORHOOD CONSTRUCTION FOR LOCAL SHAPE FEATURES.

	SJTU [18]			WPC [19]			ICIP [37]			LS-PCQA [38]			MJ-PCCD [39]		
	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
KNN	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651
Radius	0.880	0.873	1.154	0.826	0.817	12.911	0.964	0.954	0.302	0.589	0.566	0.672	0.838	0.753	0.638

as PointPCA [9], FRSVR [10], and SGW-PCQA+ [11] are analyzed as comparison methods. Then, we validate the effectiveness of the proposed two-stage fusion strategy. Specifically, we evaluate: One-stage Uniform Fusion equally averages all cues in a single step; One-stage Adaptive Fusion combines all modalities in a single step using adaptive weights; Two-stage Uniform Fusion adopts the proposed two-stage fusion, but all weights are fixed and uniform; Two-stage Adaptive Fusion uses the proposed two-stage organization with adaptive weights.

As shown in Table IV, two-stage fusion strategies consistently outperform their one-stage counterparts on both datasets. Furthermore, the proposed two-stage adaptive fusion achieves the best or tied-best overall performance across the evaluation metrics. Compared with uniform fusion, the performance gain of adaptive fusion indicates that adaptive weighting is most effective when integrated into the two-stage fusion.

The improvement from “Only 3D” to multimodal fusion confirms that perceptual quality cannot be fully explained by intrinsic 3D fidelity alone. The gains obtained by adding saliency indicate that attention-related representations provide complementary information, especially when distortions occur in visually important regions. Moreover, the superiority of two-stage fusion over one-stage fusion supports the proposed cue organization: intrinsic structure and appearance cues are first combined into a 3D fidelity representation, which is then modulated by saliency. This result is consistent with the Marr-inspired view that visual quality prediction benefits from separating the construction of representations from attentional modulation.

1) *Neighborhood Construction*: In the local shape feature extraction (Eq. 1), the neighborhood of each point can be constructed using either KNN-based or radius-based strategies. To analyze their influence, we conduct an ablation study and report the results in Table V. The KNN-based strategy generally achieves more stable and competitive performance across different datasets, indicating better consistency with subjective perception. The radius-based strategy shows comparable performance on several datasets; however, its overall improvement is limited. In addition to prediction performance, computational efficiency is also an important consideration for large-scale point clouds. In our implementation, radius-based neighborhood construction requires substantially higher neighbor-search cost, whereas KNN-based neighborhood search can be efficiently implemented using KD-tree structures. The average runtime of the radius-based strategy is approximately $16\times$ higher than that of the KNN-based strategy on the SJTU dataset. Considering both prediction accuracy and computational efficiency, the KNN-based neighborhood construction strategy is adopted in MOSAIC.

2) *Color Space*: In Sec. III-B2, the RGB color space is directly adopted considering its consistency with the stored color attributes of point cloud datasets. To further investigate the influence of color representation, we additionally evaluate MOSAIC under different color spaces, including YUV and CIE-LAB (LAB). The corresponding results are reported in Table VI. RGB achieves the best overall performance on most datasets, except on LS-PCQA. Although YUV and LAB obtain slightly better results on LS-PCQA, their improvements are marginal and accompanied by noticeable performance degradation on other datasets. This observation indicates that directly modeling color distortions in RGB space is more robust and generalizable for point cloud quality assessment. Therefore, RGB is adopted in MOSAIC.

3) *Parameter Sensitivity*: To examine whether the fixed parameters in MOSAIC introduce dataset-specific tuning, we conduct a one-at-a-time sensitivity analysis for K_g , K_p , K_c , and α , as reported in Table VII. The default setting remains competitive across all datasets, indicating that the proposed metric is not highly sensitive to moderate parameter variations. For the geometry neighborhood size K_g , reducing it to 10 or 50 causes only limited performance fluctuations on most datasets, suggesting that the structural representation is stable once sufficient local geometry is captured. Increasing the density neighborhood size K_p improves MJ-PCCD but degrades SJTU, WPC, and LS-PCQA, indicating that overly large density neighborhoods may smooth fine-grained sampling changes. A similar trend is observed for the color neighborhood size K_c , where larger neighborhoods tend to dilute local chromatic differences and reduce performance. The nonlinear mapping parameter α also shows stable behavior: $\alpha = 3$ slightly improves LS-PCQA, while the default $\alpha = 1$ provides the most balanced overall performance. Therefore, the default parameters are adopted to preserve cross-dataset robustness without per-dataset tuning.

C. Distortion-wise Performance Analysis

To further evaluate the robustness of the proposed metric under different distortion conditions, we conduct a distortion-wise performance analysis by integrating distorted samples from the SJTU [18], WPC [19], ICIP [37], and MJ-PCCD [39] datasets. Five representative distortion types are considered. Specifically, G-PCC Octree Compression includes 182 distorted point clouds, with 80 samples from WPC, 30 from ICIP, and 72 from MJ-PCCD. G-PCC Trisoup Compression includes 270 distorted point clouds, with 240 samples from WPC and 30 from ICIP. V-PCC Compression includes 279 distorted point clouds, with 180 samples from WPC, 30 from ICIP, and 69 from MJ-PCCD. Downsampling includes 114 distorted point clouds, with 54 samples from SJTU and 60 from WPC.

TABLE VI
PERFORMANCE WITH VARIOUS COLOR SPACES. THE BEST RESULT IN EACH COLUMN IS MARKED IN **BOLD**.

	SJTU [18]			WPC [19]			ICIP [37]			LS-PCQA [38]			MJ-PCCD [39]		
	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
RGB	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651
YUV	0.882	0.849	1.142	0.799	0.786	13.775	0.940	0.934	0.387	0.651	0.638	0.632	0.781	0.734	0.731
LAB	0.856	0.817	1.256	0.756	0.741	15.001	0.882	0.890	0.534	0.640	0.645	0.640	0.741	0.676	0.785

TABLE VII
PARAMETER SENSITIVITY ANALYSIS. THE DEFAULT SETTING IS $K_g = 100$, $K_p = 10$, $K_c = 10$, AND $\alpha = 1$.

	SJTU [18]			WPC [19]			ICIP [37]			LS-PCQA [38]			MJ-PCCD [39]		
	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
Default	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651
$K_g = 10$	0.892	0.882	1.096	0.846	0.837	12.231	0.969	0.964	0.279	0.628	0.604	0.648	0.807	0.729	0.691
$K_g = 50$	0.912	0.896	0.997	0.842	0.832	12.364	0.969	0.962	0.281	0.624	0.601	0.650	0.825	0.745	0.661
$K_p = 50$	0.902	0.890	1.047	0.832	0.822	12.726	0.968	0.958	0.285	0.614	0.586	0.657	0.865	0.774	0.587
$K_p = 100$	0.901	0.889	1.053	0.831	0.822	12.752	0.968	0.957	0.285	0.609	0.582	0.660	0.871	0.775	0.576
$K_c = 50$	0.897	0.876	1.074	0.842	0.835	12.376	0.960	0.947	0.319	0.614	0.600	0.657	0.815	0.754	0.678
$K_c = 100$	0.876	0.849	1.170	0.834	0.828	12.642	0.952	0.942	0.349	0.606	0.596	0.661	0.801	0.744	0.700
$\alpha = 0.5$	0.912	0.898	0.994	0.844	0.834	12.285	0.965	0.955	0.297	0.607	0.586	0.661	0.830	0.754	0.652
$\alpha = 3$	0.912	0.898	0.994	0.844	0.834	12.303	0.965	0.955	0.297	0.640	0.612	0.639	0.831	0.754	0.651

TABLE VIII
PERFORMANCE EVALUATION ON DIFFERENT DISTORTION TYPES. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN **RED** AND **GREEN**.

	G-PCC Octree			G-PCC Trisoup			V-PCC			Downsampling			Gaussian Noise		
	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
M-p2po [2]	0.489	0.173	0.220	0.531	0.511	0.208	0.508	0.487	0.181	0.087	0.713	0.286	0.147	0.145	0.210
M-p2pl [3]	0.498	0.176	0.219	0.517	0.497	0.210	0.558	0.544	0.174	0.087	0.649	0.286	0.147	0.041	0.210
PSNRyuv [4]	0.700	0.675	0.180	0.618	0.621	0.193	0.464	0.382	0.186	0.631	0.631	0.223	0.729	0.682	0.145
PCQM [5]	0.886	0.838	0.117	0.791	0.792	0.150	0.718	0.698	0.146	0.886	0.840	0.133	0.893	0.869	0.096
PointSSIM [20]	0.449	0.134	0.225	0.567	0.562	0.203	0.606	0.592	0.167	0.714	0.701	0.201	0.747	0.710	0.141
GraphSIM [6]	0.830	0.781	0.141	0.785	0.785	0.152	0.499	0.436	0.182	0.951	0.905	0.089	0.876	0.853	0.102
MPED [21]	0.535	0.494	0.213	0.548	0.545	0.206	0.688	0.696	0.153	0.933	0.903	0.103	0.855	0.820	0.110
PQI [11]	0.776	0.743	0.159	0.737	0.742	0.166	0.582	0.560	0.171	0.948	0.913	0.091	0.916	0.901	0.085
TCDM [22]	0.686	0.619	0.183	0.846	0.845	0.121	0.744	0.742	0.141	0.943	0.894	0.096	0.897	0.884	0.094
PHM [23]	0.704	0.667	0.179	0.874	0.869	0.120	0.581	0.535	0.171	0.950	0.908	0.090	0.929	0.913	0.079
MOSAIC	0.798	0.766	0.152	0.819	0.816	0.141	0.781	0.769	0.131	0.948	0.909	0.092	0.906	0.872	0.090

Gaussian Noise includes 234 distorted point clouds, with 54 samples from SJTU and 180 from WPC.

The quantitative results are reported in Table VIII. Different FR-PCQA metrics show complementary strengths under different distortion mechanisms. PCQM performs best on G-PCC Octree, PHM is particularly strong on G-PCC Trisoup and Gaussian noise, and GraphSIM and PQI are competitive for downsampling. In contrast, MOSAIC achieves the best performance on V-PCC compression with (PLCC / SROCC / RMSE) values of (0.781 / 0.769 / 0.131), outperforming both TCDM and PHM. This advantage suggests that combining intrinsic 3D geometry/color fidelity with projected saliency consistency is helpful for compression artifacts whose perceptual impact depends on both local 3D structure and view-dependent appearance. Although MOSAIC is not always the best on every distortion type, it remains competitive across all categories without distortion-specific switching or supervised fitting, demonstrating stable generalization for heterogeneous point cloud degradations.

D. Discussion on Recent FR-PCQA

The taxonomy in Sec. II shows that FR-PCQA methods differ not only in feature design, but also in how quality evidence is organized. PCQM and PQI represent the local-structure lineage: PCQM selects compact geometry/color descriptors from matched neighborhoods, whereas PQI improves key-point coverage and neighborhood control through multiscale

neighbor-to-neighbor comparison. TCDM and PHM represent a more recent perceptual-hypothesis-driven lineage: TCDM models quality through transformational complexity, whereas PHM switches between visible-difference and appearance-degradation strategies according to distortion degree. Table IX summarizes these technical distinctions and compares them with MOSAIC. Regression-based and learning-based methods are discussed separately because their scores can depend on training data, split protocols, or learned backbones.

This distinction also provides a useful criterion for evaluating MOSAIC. The contribution of recent non-learning metrics is not merely the use of handcrafted features; rather, strong metrics identify a concrete weakness of prior FR-PCQA, propose a testable perceptual or computational explanation, and validate it with broad cross-dataset comparisons and targeted analyses. PQI links robustness to multiscale key-point coverage and controlled neighborhood size, TCDM links quality to the complexity of recovering the reference from the distorted point cloud, and PHM links prediction behavior to distortion-level-dependent HVS strategies. Following this standard, the central claim of MOSAIC is that FR-PCQA benefits from multimodal perceptual cue fusion with a two-stage organization: native 3D structure and color form an intrinsic fidelity estimate, while projected saliency provides attention-related modulation. The ablation in Table IV, the neighborhood and color-space analyses in Tables V and VI, and the distortion-wise results in Table VIII are therefore

TABLE IX
TECHNICAL COMPARISON OF REPRESENTATIVE FR-PCQA METRICS.

Metric	Technical focus	Cue extraction and fusion	Main advantages	Main limitations
PCQM [5]	Intrinsic 3D geometry and color quality measurement.	Extracts local geometric and color descriptors from matched point neighborhoods and combines them into a handcrafted full-reference quality score.	Training-free, relatively efficient, and strong for several compression-related distortions; performs best on G-PCC Octree in Table VIII.	Mainly relies on local 3D geometry/color discrepancies; does not explicitly model projected attention or view-dependent saliency; performance drops on heterogeneous large-scale databases such as LS-PCQA and MJ-PCCD in Table I.
PQI [1]	Multiscale key-point and neighborhood quality indexing.	Selects multiscale key points and measures neighbor-to-neighbor geometry and attribute similarity around them.	Improves robustness over single-scale point-wise metrics; strong on SJTU and ICIP and competitive for downsampling/Gaussian-noise categories.	Still depends on key-point and neighborhood construction; mainly describes local geometry/attribute consistency and does not use an explicit attention-related cue; less effective on LS-PCQA and MJ-PCCD in our comparison.
TCDM [22]	Transformational-complexity modeling inspired by predictive coding.	Uses 3D Voronoi partitioning to form local patch pairs, then applies a space-aware vector autoregressive model to compute self-complexity, transformational complexity, and prediction terms.	Provides a principled patch-level transformation view; obtains the best results on SJTU and MJ-PCCD in Table I.	Sensitive to patch partitioning, autoregressive neighborhood scale, and local point-distribution stability; reported V-PCC behavior can vary with point-cloud sparsity; weaker than MOSAIC on WPC, LS-PCQA, and V-PCC.
PHM [23]	Perception-guided hybrid modeling for different distortion levels.	Combines a visible-difference branch for high-quality samples using texture-complexity-compensated PSNR _Y and an appearance-degradation branch using graph signal variation and spectral graph wavelet statistics; fuses them with a distortion-degree-dependent nonlinear weight.	Strong for ICIP, G-PCC Trisoup, and Gaussian noise; explicitly considers that HVS strategies may differ between subtle and severe distortions.	Relatively complex graph/spectral pipeline and explicit high/low-quality decomposition; highest runtime in Table III; weaker than MOSAIC on LS-PCQA and V-PCC.
MOSAIC	Multimodal perceptual cue fusion with two-stage organization.	Measures 3D geometry fidelity, 3D color fidelity, and 2D saliency fidelity, then first fuses intrinsic 3D cues and subsequently incorporates projected saliency through adaptive bounded weighting.	Training-free and interpretable; integrates native 3D structural/appearance evidence with view-based attention-related evidence; best on LS-PCQA and V-PCC and competitive across datasets and distortion types.	Uses fixed six-view saliency projection and fixed neighborhood parameters; does not yet model interactive viewing trajectories.

intended to test the cue-organization hypothesis rather than only report another feature combination.

TCDM is technically different from MOSAIC in both representation and pooling philosophy. It first samples seed points from the reference point cloud, constructs a 3D Voronoi partition to form local reference-distorted patch pairs, and then applies a space-aware vector autoregressive model to perform self-prediction and cross-prediction of geometry and color channels. The quality score is derived from self-complexity, transformational complexity, and prediction-related terms. This predictive-coding view is powerful, as reflected by its leading performance on SJTU and MJ-PCCD in Table I. However, TCDM remains a local patch-prediction metric: its behavior depends on patch partitioning, autoregressive neighborhood scale, and the stability of local point distributions. The TCDM paper also reports inconsistent V-PCC behavior across databases, where V-PCC may change the point count differently depending on the sparsity of the reference point cloud. This limitation is visible in our distortion-wise evaluation: MOSAIC outperforms TCDM on V-PCC compression, with PLCC/SROCC/RMSE of 0.781/0.769/0.131 versus 0.744/0.742/0.141. MOSAIC also gives stronger WPC and LS-PCQA results in Table I. These gains suggest that measuring explicit geometry, color, and projected saliency fidelity can be more stable than relying only on local transformational complexity when distortions alter both 3D sampling and view-dependent appearance.

PHM is also related, but its technical design follows a different hypothesis. It divides quality prediction into two perception-inspired branches: a visible-difference branch for high-quality samples, where PSNR_Y is compensated by texture complexity estimated from an autoregressive model, and an appearance-degradation branch for low-quality samples, where local graph construction, geometry signal variation, spectral graph wavelet coefficients, and weighted co-occurrence statistics are used to describe geometry and texture degradation. The two components are then combined by a distortion-degree-dependent nonlinear weighting function. This hybrid design explains PHM's strong performance on ICIP, G-PCC Trisoup, and Gaussian-noise distortions. Its limitation is that the method depends on a relatively elaborate graph/spectral pipeline and an explicit high-quality/low-quality decomposition. As shown in Table III, PHM requires 14.96 seconds per distorted point cloud on average, compared with 6.41 seconds for MOSAIC. PHM is also less effective on LS-PCQA and V-PCC in our evaluation. In particular, MOSAIC improves LS-PCQA from 0.546/0.500/0.697 to 0.622/0.598/0.652 and V-PCC from 0.581/0.535/0.171 to 0.781/0.769/0.131. Therefore, the advantage of MOSAIC is not that it uses more complex handcrafted modules, but that it uses a single two-stage cue-fusion formulation to integrate intrinsic 3D fidelity and projected attention-related fidelity without distortion-severity branch selection.

Regression-based FR-PCQA methods form a separate com-

parison category rather than ablation variants. PointPCA [9] learns from PCA-based local descriptors, FRSVR [10] integrates five full-reference distortion measures through SVR, SGW-PCQA+ [11] uses spectral graph wavelet representations with supervised score integration, and FSM [24] selects features from existing metrics before ridge or SVR-based prediction. These methods show the benefit of subjective-label score fitting, but their performance also reflects feature selection, training/test splits, and learned regression functions in addition to perceptual cue design. For this reason, they are summarized separately in Table II rather than mixed with the training-free metrics in Table I.

Deep-learning FR-PCQA is another active line. Tliba *et al.* [28] learn local deep features from reference and distorted point-cloud patches and compare them via feature cross-correlation. Projection-based deep baselines render point clouds into multiple views, extract visual features using ConvNeXt or Swin Transformer, and regress feature similarities into quality scores [27]. CT-PCQA [12] combines CNN and Transformer backbones on projected views, while recent MLLM-based FR-PCQA uses rendered views, prompt-based responses, fine-tuned absolute-quality scoring, and SVR-based score integration [13]. Adjacent NR-PCQA work also shows a similar multimodal trend, with LMM-PCQA [29] combining LMM quality-token outputs with multiscale structural features. These methods can learn high-level visual representations, but their practical reliability often depends on subjective-label supervision, split design, backbone or prompt choice, rendering settings, and retraining or fine-tuning when the database changes. When complete directly comparable numerical results are not available, we discuss such methods qualitatively rather than adding partial numbers to the comparison table.

Compared with regression-based and deep-learning models, MOSAIC is training-free, interpretable, and does not require MOS-supervised feature learning, quality-token fine-tuning, or SVR-based score integration; the logistic mapping is used only for standard evaluation. Its components correspond to visible perceptual cues, including local 3D structure, local point density, local color contrast, and multi-view saliency consistency. The resulting metric is therefore easier to reproduce and less dependent on training data while still combining native 3D evidence with projected attention-related evidence.

E. Limitations and Validity Considerations

Several validity issues should be noted. First, Table I and Table II intentionally separate training-free metrics from supervised regression or learning-based references. This separation is useful for positioning MOSAIC within recent FR-PCQA research, but the supervised-reference table should not be read as a homogeneous leaderboard because its numbers may depend on dataset-specific score fitting, learned backbones, or challenge protocols. Second, paired prediction scores are not available for all recently reported methods, which limits the possibility of applying statistical significance tests uniformly across all baselines. We therefore emphasize consistent trends across datasets and distortion types rather than isolated marginal differences.

Third, the saliency branch uses six fixed orthogonal projections. This design keeps the metric deterministic and computationally tractable, but it does not model user-dependent viewing trajectories, dynamic head motion, or occlusion changes in interactive rendering. As a result, MOSAIC may be less sensitive to distortions that are perceptually important only under specific viewing conditions. Fourth, all neighborhood sizes and fusion parameters are fixed across datasets. This supports reproducibility and reduces the risk of dataset-specific tuning, but it may not be optimal for every point density, content category, or acquisition pipeline. These limitations suggest two future directions: view-adaptive or 3D saliency modeling, and more efficient implementations that preserve the current training-free and interpretable design.

V. CONCLUSION

In this work, we proposed MOSAIC, a Multimodal Perceptual Cue Fusion framework for full-reference point cloud quality assessment. From Marr's three-level perspective, MOSAIC represents point cloud degradation through complementary perceptual fidelity cues from 3D geometry, 3D color, and 2D saliency, corresponding to structural consistency, appearance consistency, and attention-related visual importance. A two-stage adaptive fusion strategy first integrates intrinsic 3D fidelity cues and then incorporates saliency fidelity to predict the final perceptual fidelity score without MOS-supervised training. Experiments on five public PCQA datasets demonstrate that MOSAIC achieves competitive performance and robust generalization across diverse distortion types. Future work will explore more efficient implementations, view-adaptive saliency modeling, and broader validation under interactive viewing conditions while preserving the training-free and interpretable nature of the framework.

REFERENCES

- [1] J. Zhang, T. Chen, D. Ding, and Z. Ma, "Revisit point cloud quality assessment: Current advances and a multiscale-inspired approach," *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 10, pp. 8886–8899, 2025.
- [2] R. Mekuria, Z. Li, C. Tulvan, and P. Chou, "Evaluation criteria for PCC (Point Cloud Compression)," *ISO/IEC MPEG n16332*, Feb. 2016.
- [3] D. Tian, H. Ochimizu, C. Feng, R. Cohen, and A. Vetro, "Geometric distortion metrics for point cloud compression," in *IEEE International Conference on Image Processing (ICIP)*, 2017.
- [4] E. M. Torlig, E. Alexiou, T. A. Fonseca, R. L. de Queiroz, and T. Ebrahimi, "A novel methodology for quality assessment of voxelized point clouds," in *Applications of Digital Image Processing XLI*, vol. 10752. SPIE, 2018, pp. 174–190.
- [5] G. Meynet, Y. Nehmé, J. Digne, and G. Lavoué, "PCQM: A full-reference quality metric for colored 3d point clouds," in *Twelfth International Conference on Quality of Multimedia Experience (QoMEX)*, 2020.
- [6] Q. Yang, Z. Ma, Y. Xu, Z. Li, and J. Sun, "Inferring point cloud quality via graph similarity," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 6, pp. 3015–3029, 2022.
- [7] Y. Zhang, Q. Yang, and Y. Xu, "MS-GraphSIM: Inferring point cloud quality via multiscale graph similarity," in *Proceedings of the 29th ACM International Conference on Multimedia*, 2021, pp. 1230–1238.
- [8] Z. Wang, Y. Zhang, Q. Yang, Y. Xu, J. Sun, and S. Liu, "Point cloud quality assessment using 3d saliency maps," *arXiv preprint arXiv:2209.15475*, 2022.
- [9] E. Alexiou, X. Zhou, I. Viola, and P. Cesar, "PointPCA: point cloud objective quality assessment using PCA-based descriptors," *Journal on Image and Video Processing*, vol. 2024, no. 1, Aug. 2024.

- [10] R. Watanabe, S. N. Sridhara, H. Hong, E. Pavez, K. Nonaka, T. Kobayashi, and A. Ortega, "Full reference point cloud quality assessment using support vector regression," *Signal Processing: Image Communication*, vol. 131, p. 117239, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0923596524001401>
- [11] R. Watanabe, K. Nonaka, E. Pavez, T. Kobayashi, and A. Ortega, "Full-reference point cloud quality assessment using spectral graph wavelets," in *IEEE International Conference on Image Processing (ICIP)*, 2024, pp. 3313–3319.
- [12] Y. Zhou, Z. Zhang, W. Sun, X. Min, and G. Zhai, "CT-PCQA: A convolutional neural network and transformer combined method for point cloud quality assessment," *Signal Processing: Image Communication*, vol. 138, p. 117371, 2025.
- [13] R. Watanabe, T. Konno, H. Sankoh, B. Tanaka, and T. Kobayashi, "Full-reference point cloud quality assessment with multimodal large language models," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [14] D. Marr, *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman, 1982.
- [15] M. S. Landy, L. T. Maloney, E. B. Johnston, and M. Young, "Measurement and modeling of depth cue combination: In defense of weak fusion," *Vision Research*, vol. 35, no. 3, pp. 389–412, 1995.
- [16] D. C. Knill and J. A. Saunders, "Do humans optimally integrate stereo and texture information for judgments of surface slant?" *Vision Research*, vol. 43, no. 24, pp. 2539–2558, 2003.
- [17] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 20, no. 11, pp. 1254–1259, 1998.
- [18] Q. Yang, H. Chen, Z. Ma, Y. Xu, R. Tang, and J. Sun, "Predicting the perceptual quality of point cloud: A 3D-to-2D projection-based exploration," *IEEE Transactions on Multimedia*, vol. 23, pp. 3877–3891, 2021.
- [19] Q. Liu, H. Su, Z. Duanmu, W. Liu, and Z. Wang, "Perceptual quality assessment of colored 3D point clouds," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 8, pp. 3642–3655, 2023.
- [20] E. Alexiou and T. Ebrahimi, "Towards a point cloud structural similarity metric," in *IEEE International Conference on Multimedia & Expo Workshops (ICMEW)*, 2020.
- [21] Q. Yang, Y. Zhang, S. Chen, Y. Xu, J. Sun, and Z. Ma, "MPED: Quantifying point cloud distortion based on multiscale potential energy discrepancy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, pp. 6037–6054, 2022.
- [22] Y. Zhang, Q. Yang, Y. Zhou, X. Xu, L. Yang, and Y. Xu, "TCMD: Transformational complexity based distortion metric for perceptual point cloud quality assessment," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 10, pp. 6707–6724, 2024.
- [23] Y. Zhang, Q. Yang, Y. Xu, and S. Liu, "Perception-guided quality metric of 3d point clouds using hybrid strategy," *IEEE Transactions on Image Processing*, vol. 33, pp. 5755–5770, 2024.
- [24] J. Prazeres, R. Rodrigues, M. Pereira, and A. M. G. Pinheiro, "Point cloud objective quality: Benchmarking features and quality evaluation," *arXiv preprint arXiv:2504.03381*, 2025.
- [25] X. Wu, Y. Zhang, C. Fan, J. Hou, and S. Kwong, "Subjective quality database and objective study of compressed point clouds with 6DoF head-mounted display," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 12, pp. 4630–4644, 2021.
- [26] A. Javaheri, C. Brites, F. Pereira, and J. Ascenso, "Joint geometry and color projection-based point cloud quality metric," *IEEE Access*, vol. 10, pp. 90 481–90 497, 2022.
- [27] Z. Zhang, Y. Zhou, W. Sun, X. Min, and G. Zhai, "Simple baselines for projection-based full-reference and no-reference point cloud quality assessment," *arXiv preprint arXiv:2310.17147*, 2023.
- [28] M. Tliba, A. Chetouani, G. Valenzise, and F. Dufaux, "Point cloud quality assessment using cross-correlation of deep features," in *Proceedings of the 2nd Workshop on Quality of Experience in Visual Multimedia Applications*, 2022, pp. 63–68.
- [29] Z. Zhang, H. Wu, Y. Zhou, C. Li, W. Sun, C. Chen, X. Min, X. Liu, W. Lin, and G. Zhai, "LMM-PCQA: Assisting point cloud quality assessment with LMM," *arXiv preprint arXiv:2404.18203*, 2024.
- [30] H. Hoppe, T. DeRose, T. Duchamp, J. McDonald, and W. Stuetzle, "Surface reconstruction from unorganized points," in *Proceedings of the 19th Annual Conference on Computer Graphics and Interactive Techniques*, 1992, pp. 71–78.
- [31] H. Knutsson, C.-F. Westin, and M. Andersson, "Representing local structure using tensors ii," in *Image Analysis: 17th Scandinavian Conference*. Springer, 2011, pp. 545–556.
- [32] W. Zhou, G. Yue, R. Zhang, Y. Qin, and H. Liu, "Reduced-reference quality assessment of point clouds via content-oriented saliency projection," *IEEE Signal Processing Letters*, vol. 30, p. 354–358, 2023.
- [33] H. R. Tavakoli and J. Laaksonen, "Bottom-up fixation prediction using unsupervised hierarchical models," in *Asian Conference on Computer Vision*. Springer, 2016, pp. 287–302.
- [34] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [35] M. O. Ernst and M. S. Banks, "Humans integrate visual and haptic information in a statistically optimal fashion," *Nature*, vol. 415, no. 6870, pp. 429–433, 2002.
- [36] L. N. Vinke, I. M. Bloem, and S. Ling, "Saturating nonlinearities of contrast response in human visual cortex," *The Journal of Neuroscience*, vol. 42, no. 7, pp. 1292–1302, 2022. [Online]. Available: <https://doi.org/10.1523/JNEUROSCI.0106-21.2021>
- [37] S. Perry, H. P. Cong, L. A. da Silva Cruz, J. Prazeres, M. Pereira, A. Pinheiro, E. Dumic, E. Alexiou, and T. Ebrahimi, "Quality evaluation of static point clouds encoded using MPEG codecs," in *IEEE International Conference on Image Processing (ICIP)*. IEEE, 2020, pp. 3428–3432.
- [38] Y. Liu, Q. Yang, Y. Xu, and L. Yang, "Point cloud quality assessment: Dataset construction and learning-based no-reference metric," *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 19, no. 2s, pp. 1–26, 2023.
- [39] D. Lazzarotto, M. Testolina, and T. Ebrahimi, "Subjective performance evaluation of bitrate allocation strategies for MPEG and JPEG Pleno point cloud compression," *Journal of Image and Video Processing*, vol. 2024, no. 14, 2024. [Online]. Available: <https://doi.org/10.1186/s13640-024-00629-0>
- [40] J. Antkowiak, J. Baina, V. Baroncini, N. Chateau, A. C. F. Pessoa, S. Colonnese, L. Contin, and J. Caviedes, "Final report from the video quality experts group on the validation of objective models of video quality assessment," Video Quality Experts Group (VQEG), Tech. Rep., Mar. 2000.